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# TME–602

## B. TECH. (MECH.) (SIXTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

DESIGN OF MACHINE ELEMENTS—II

**Time : Three Hours**

**Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is nipping in leaf springs ? Explain with mathematical expressions. (CO1)

(b) A cylinder relief valve should blow at a pressure of 1.5 N/mm² and should lift by 6 mm for a 5 per cent increase in pressure. The diameter of the valve is 60 mm. Design the valve spring giving a suitable combination of (a) mean coil diameter, (b) wire diameter, (c) number of active and inactive coils. Permissible shear stress for the spring material may be taken as 500 N/mm² and $G = 0.85 \times 10^5$ MPa. (CO1)

(c) A semi-elliptical laminated spring is to carry a load of 5000 N and consists 8 leaves 46 mm of the leaves being of full length. The spring is to be made 1000 mm between the eyes and is held at the centre by a 60 mm wide band. Assume that the spring is initially stressed so as to induce an equal stress of 500 N/mm² when fully loaded. Design the spring giving :

(i) thickness of leaves

(ii) maximum deflection and camber.

Assume $E = 2.1 \times 10^5$ N/mm².

2. (a) (i) What are the important factors considered for the selection of flat belt for power transmission ?

(ii) Explain the advantages and disadvantages of V-belt drive over Flat belt drive. (CO2)

(b) A V-belt drive is to transmit 14.7 kW to a compressor. The motor speed is 1150 rev/min and the compressor pulley runs at 400 rev/min. Determine the size and number of belts required. (CO2)

(c) It is required to design a chain drive to connect a 12 kW, 1400 rpm electric motor to a centrifugal pump running at 700 rpm. The service conditions involve moderate shocks.

(i) Select a proper roller chain and give a list of its dimensions.

(ii) Determine the pitch circle diameters of driving and driven sprockets.

(iii) Determine the number of chain links. (CO2)

| **Pitch (mm)** | **Weight of chain (N/m)** | | |
|---|---|---|---|
| | Simplex | Duplex | Triplex |
| 9.525 | 4.1 | 7.7 | 10.9 |
| 12.7 | 7.1 | 13.6 | 20.2 |
| 15.785 | 9.1 | 18.2 | 27.3 |
| 19.05 | 11.7 | 23.6 | 35.4 |

3. (a) Explain the concept of the formative number of teeth in helical gears with a neat sketch and mathematical expressions. (CO3)

(b) A compressor running at 300 rev/min is driven by a 15 kW, 1200 rev/min motor through a 14 full depth gears. The centre distance is 0.375 m. The motor pinion is to be of C-30 forged steel hardened and tempered, and the driven gear is to be of cast steel. Assuming medium shock condition : (a) Determine the module, the face width, and the number of teeth on each gear. (b) Check the gears for dynamic load and wear. (CO3)

(c) Two helical gears are used in a speed reducer that is to be driven by an internal combustion engine. The rated power of the speed reducer is 75 kW at a pinion speed of 1200 rev/min. The speed ratio is 3 to 1. Assume medium shock conditions and 24 hr operation.

Determine the module, face width, number of teeth in each gear, check for dynamic and wear load if the teeth are 20° full depth in the normal plane. (CO3)

4. (a) Explain the concept of the formative number of teeth in bevel gears with a neat sketch and mathematical expressions. (CO4)

(b) Design a bevel gear drive between two shafts whose axes are at right angles. Speed of pinion shaft is 240 rev/min and that of the gear shaft is 120 rev/min. Pinion is to have 21 teeth of involute profile with module of 20 mm and a pressure angle of 20° and is to be of suitable material. Gear is of cast iron. Power at gear shaft = 75 (CO4)

(c) Design (without checking for dynamic and wear load) the 20 degree involute worm and worm gear which will transmit 11.25 kW between shafts that are 0.25 m apart, if the speed reduction is to be 10.5 to 1, and the driving shaft is turning at 1200 rev/min. (CO4)

5. (a) Explain the various required properties of sliding contact bearing material in brief for its selection. (CO5, CO6)

(b) A journal bearing is proposed for a centrifugal pump. The diameter of the journal is 0.15 m and the load on it is 40 kN and its speed is 900 rev/min. Complete the design calculation for the bearing. (CO5, CO6)

(c) Explain with reference to a neat plot the importance of the bearing characteristic curve. (CO5, CO6)



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# TME–606

## B. TECH. (ME) (SIXTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

COMPUTER AIDED MANUFACTURING

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) List advantages and limitations of NC and CNC machine tools over conventional machine tools. (CO1, CO2)

   (b) Detail elements of Industry 4.0. Explain various stages of growth in industrial revolution. (CO1, CO2)

   (c) Enumerate basic elements of CAM. Compare CAM against conventional manufacturing process. (CO1, CO2)

2. (a) Specify various features of DNC systems. Explain general software features available in DNC communications. (CO3)

*P. T. O.*

(b) Compare "Distributed Numeric Control" and "Direct Numeric Control" Systems. Explain various network topologies. (CO3)

(c) List various features of DNC software. Explain DNC communication with listing of advantages of DNC systems. (CO3)

3. (a) Enumerate Computer Aided Process Planning Methods. Compare Manual and Computer Aided Process Planning. (CO4, CO5)

(b) Analyze Variant, Generative and Automated Process Planning Methods with detailed explanation. (CO4, CO5)

(c) Explain "Group Technology (GT)", "Flexible Manufacturing System (FMS)" and "Manufacturing Execution System (MES)" in detail.

(CO4, CO5)

4. (a) Classify Automation. Compare Open and Closed loop systems in CAM.

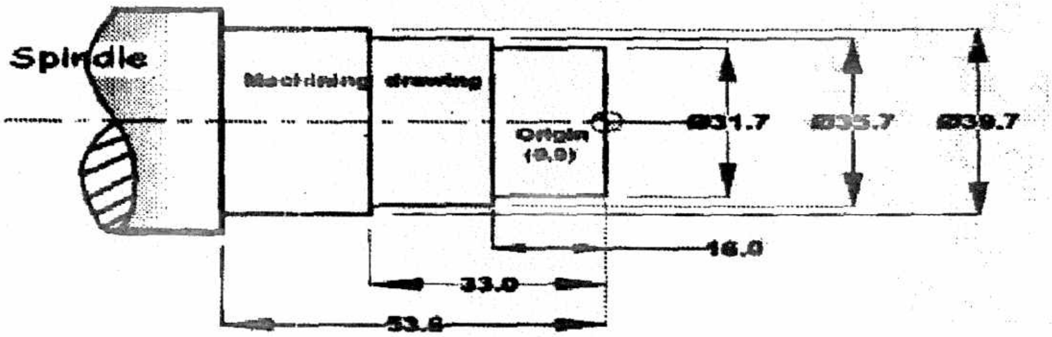
(CO2, CO6)

(b) Differentiate between ordinary and NC Machine tools. Elaborate various methods for improving accuracy and productivity. (CO2, CO6)

(c) Classify various parts of an NC Program. Explain generation of NC Programmes through CAD/CAM systems for a sample geometry.

(CO2, CO6)

5. (a) Write the APT geometry NC part program for defining the part outline for turning operation (Fig.1). (CO6)




(b) Explain Computer aided NC Programming in APT language. Specify Manual Programming for simple parts, e.g., turning, milling, drilling. (CO6)

(c) Elaborate various elements of Expert Systems and Artificial Intelligence in manufacturing with various applications of Neural Networks in manufacturing. (CO6)



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# TME–609

## B. TECH. (SIXTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023
### INDUSTRIAL ENGINEERING AND PROJECT MANAGEMENT

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

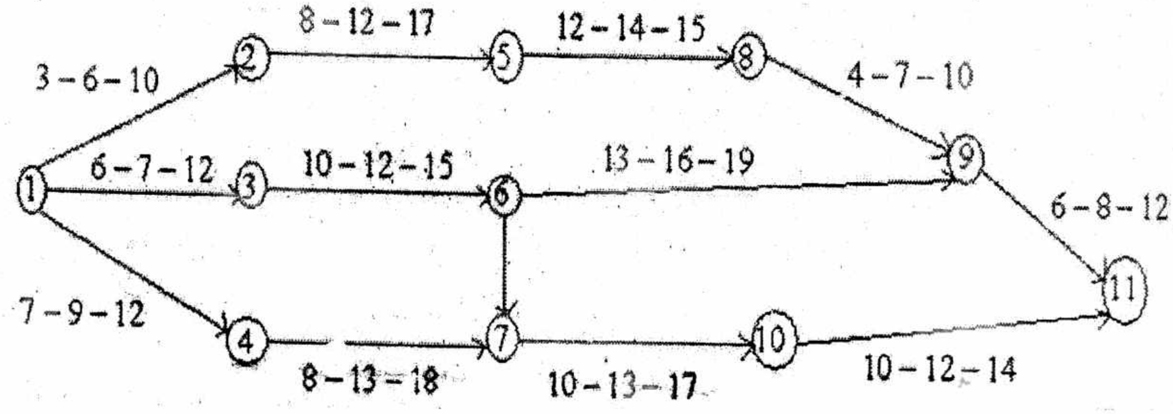
(iv) Each sub-question carries 10 marks.

1. (a) What is time study and how is it used to measure and improve productivity in various industries ? What are the steps involved in conducting a time study to analyze and improve work processes ? (CO2)

(b) What is motion study and how does it contribute to improving efficiency and reducing fatigue in work processes ? How can motion study be used to identify and eliminate wasteful movements in a manufacturing or service process, and what are some common techniques and tools used in motion study analysis ? (CO2)

(c) What are the key considerations and principles involved in designing an effective plant layout that optimizes material handling efficiency and minimizes operational costs in manufacturing facilities ? (CO2)

2. (a) Calculate the variance and expected time for each activity : (CO1, CO2)




(b) A small project consisting of eight activities has the following characteristics : (CO1, CO2)

**Time-Estimates (in weeks)**

| Activity | Preceding Activity | Most Optimistic time (a) | Most likely time (m) | Most Pessimistic time (b) |
|---|---|---|---|---|
| A | None | 2 | 4 | 12 |
| B | None | 10 | 12 | 26 |
| C | A | 8 | 9 | 10 |
| D | A | 10 | 15 | 20 |
| E | A | 7 | 7.5 | 11 |
| F | B, C | 9 | 9 | 9 |
| G | D | 3 | 3.5 | 7 |
| H | E, F, G | 5 | 5 | 5 |

(i) Draw the PERT network for the project.

(ii) Prepare the activity schedule for the project.

(c) From the following particulars, find out the break-even-point : (CO1, CO2)

| | ₹ |
|---|---|
| Variable Cost per unit | 15 |
| Fixed Expenses | 54,000 |
| Selling Price per unit | 20 |

What should be the selling price per unit, if the break-even point should be brought down to 6,000 units ?

3. (a) How can inventory control be improved through the implementation of codification and standardization methods, and what are some of the benefits and challenges associated with these approaches in managing inventory levels and reducing waste in a supply chain ? (CO3, CO4)

(b) Explain the following terms : (CO3, CO4)

(i) Optimistic time

(ii) Most likely time

(iii) Pessimistic time

(iv) Expected time

(v) Variance

(c) Discuss the terms forecasting, routing, scheduling, dispatching and follow up in details with suitable examples. (CO3, CO4)

4. (a) What is project management and why is it considered a crucial discipline in achieving successful outcomes, ensuring efficient resource allocation, and mitigating risks in complex endeavors across various industries ? (CO5)

(b) What is the significance of project selection and feasibility study in project management, and what key factors are typically considered during the process to determine the viability, profitability, and potential risks of a project before its initiation ? (CO5)

(c) How does the accurate estimation of project costs and the consideration of cost of capital contribute to effective project planning and decision-making, and what are some common methods and factors involved in estimating project costs and determining the appropriate cost of capital ? (CO5)

5. (a) What is a Project Management Information System (PMIS) and how does it enhance project planning, execution, monitoring, and control ? Furthermore, what are the key features and benefits of utilizing a PMIS in managing complex projects across various industries ? (CO6)

(b) How does effective teamwork impact the success of project management, and what are some key strategies and practices project managers can employ to foster collaboration, communication, and synergy among team members in order to achieve project objectives ? (CO6)

(c) What is the role of project evaluation and auditing in project management, and how do these processes help assess the overall performance, success, and adherence to project objectives, as well as identify areas for improvement and ensure accountability throughout the project lifecycle ? (CO6)

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# TMS–601

## B. TECH. (ME) (SIXTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

AUTOMOTIVE MECHATRONICS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Draw a block diagram showing different essential components of a mechatronic system and explain it briefly. (CO1)

   (b) What do you understand by the powertrain in an automobile ? Describe with its types. (CO1)

   (c) Give different classifications of automobile based on construction and transmission. (CO1)

2. (a) Discuss about the typical sensor measurement system with a block diagram. (CO2)

   (b) Explain the working and use of a Hall effect sensor with a neat diagram. (CO2)

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(c) Discuss about the working principle of the following automobile sensors : (CO2)

(i) knock sensor and

(ii) water in fuel sensor.

3. (a) What is a solenoid ? Explain its working and uses with a diagram. (CO3)

(b) Explain the working of a brushed dc motor with a diagram. (CO3)

(c) Explain the working of an induction motor and give a relation between the synchronous and rotor speeds for the induction motor. (CO3)

4. (a) Discuss about the starting system of a car with diagram. (CO4, CO6)

(b) Write a short note on the fuel injection systems in automobile. (CO4, CO6)

(c) What is the Electric Control Unit (ECU) used in automobile ? (CO4, CO6)

5. (a) Explain the working of a series hybrid vehicle with diagram. (CO5, CO6)

(b) What is a fuel cell ? Draw the schematic diagram of a fuel cell vehicle. (CO5, CO6)

(c) Write a short note on the electric vehicles. (CO5, CO6)



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Roll No. ..............................

# TAS–601

## B. TECH. (ME) (SIXTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

AIRCRAFT PROPULSION

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the basic principle of jet propulsion ? (CO1)

   (b) Draw a Systematic diagram of a typical Gas turbine Engine and Label all the sections. (CO1)

   (c) What are the factors that affect the thrust of a propeller ? (CO1)

2. (a) Explain the importance of a nozzle in a gas turbine engine. (CO2)

   (b) What are the factorsthat affect the performance of Combustion Chamber ? Explain in detail. (CO2)

   (c) Derive an equation for the incremental velocity of rocket (Rocket Equation). State the assumptions made, if any. (CO2)

*P. T. O.*

3. (a) Explain Rankine-Froude Momentum Theory of Propulsion. (CO3)

   (b) Why is it that a more powerful engine would need more propeller blades ? (CO3)

   (c) State the difference between air breathing engine and rocket engine. (CO3)

4. (a) Why combustion chamber is used gas turbine ? (CO4)

   (b) State the three types of combustion chamber in detail. (CO4)

   (c) How are high speed propulsion systems classified ? (CO6)

5. (a) Discuss the development of modern rocketry during the post-world war era. (CO5)

   (b) Explain in detail about Solid Rocket propulsion. (CO5)

   (c) Describe the selection criteria of solid propellant grains for various grain configurations. (CO6)

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# DEDE–01

## B. TECH. (ME) (SIXTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023

DESIGN FOR SUSTAINABLE DEVELOPMENT

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Do we have anything that is fully sustainable ? Prove it with an example. (CO1, CO4)

   (b) Explain the four pillars of sustainability circle. (CO1, CO2)

   (c) What is emotionally durable design ? Explain. (CO1, CO2)

2. (a) Explain PSS design for sustainability with suitable example. (CO2, CO3)

   (b) What are the differences between classical LCA and Fast track LCA ? (CO1, CO2)

   (c) Explain cradle to gate system with any example. (CO2, CO3)

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3. (a) Explain Dematerialization approach for sustainability with example. (CO1, CO3)

   (b) Explain how the reduce approach works for sustainability with an example. (CO1, CO3)

   (c) Explain use-oriented PSS with and example. (CO1, CO3)

4. (a) Enlist the different barriers for designers in sustainable PSS. (CO1, CO2)

   (b) What are the five different stages of MSDS ? Explain. (CO1, CO3)

   (c) How we construct the system map. Explain with diagram. (CO1, CO3)

5. (a) Create and explain an empathy map for a 3D printer. (CO5)

   (b) Explain AEIOU Mapping in detail with the table. (CO1, CO4)

   (c) Is design thinking applicable for sustainability ? Explain in detail. (CO1, CO3)



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# DEDE–06

## B. TECH. (ME) (SIXTH SEMESTER)
## END SEMESTER EXAMINATION, June, 2023

MODELLING OF CONTROL SYSTEMS

**Time : Three Hours**

**Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

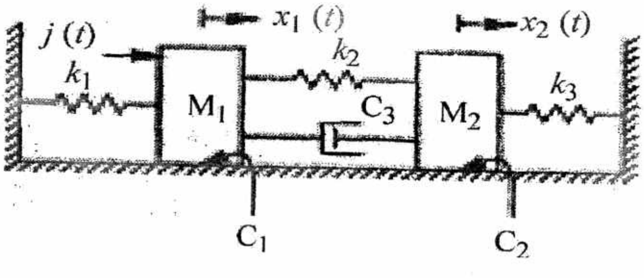
(iv) Each sub-question carries 10 marks.

(v) Answers must be accompanied by neat and labelled diagrams.

(vi) Suitable assumptions can be made for any missing data.

1. (a) Explain the applications of mechatronic systems in the automotive and aerospace domain. (CO1)

(b) Derive the equations of motion of the two mass system shown in the figure. (CO1)

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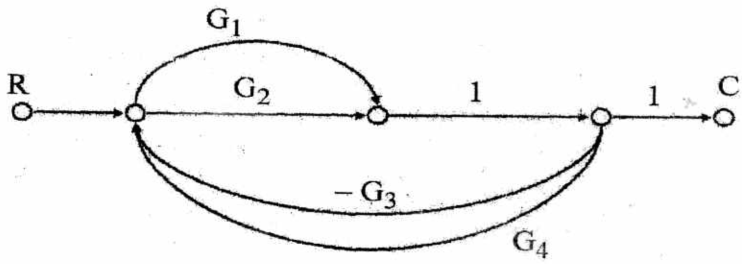
(c) Discuss about the different steps involved in the control system design. (CO1)

2. (a) Explain the principles of superposition and homogeneity with suitable examples. (CO2)

(b) Obtain the transfer function of the system who's signal flow graph is as shown below : (CO2)




(c) Find the inverse Laplace transform of $Y(s) = \dfrac{s+6}{(s+1)(s+3)}$ and sketch the solution. (CO2)

3. (a) Find the poles, residue and zeros of the transfer function given by $H(s) = \dfrac{2s+1}{s^2+5s+6}$ and also plot the poles and zeros. Give the Matlab command to find the poles, residue and zeros. (CO3)

(b) Find the transfer function of the system that is represented in the standard state space form by : (CO3)

$$A = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, C = [0 \;\; 1] \text{ and } D = [0]$$

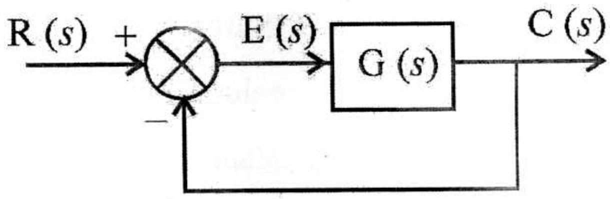
(c) Explain controllability and observability. What are the different canonical forms, and what is their significance ? (CO3)

4. (a) What is meant by type of control system ? Derive expressions for the steady state error of a type 2 control system to step, ramp and parabolic inputs. Give the Matlab commands for the same. (CO4)

(b) For the unity feedback system shown, where (CO4)

$$G(s)=\frac{450(s+8)(s+12)(s+15)}{s(s+38)(s^2+2s+28)}$$

find the steady-state errors for the following test inputs : Step, Ramp and Parabolic :




(c) Explain the concept of transient analysis. Sketch the different response of a second order system based on natural frequency and damping ratio. (CO4)

5. (a) Explain the root locus method with a relevant example. (CO5)

(b) Determine the stability of the closed-loop transfer function : (CO5)

$$T(s)=\frac{10}{s^5+2s^4+3s^3+6s^2+5s+3}$$

(c) What is PID control ? Explain the tuning of a PID controller with a suitable example. (CO5)



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Roll No. ME VIth ..............................

# DEEM–03

ME

## B. TECH. (SIXTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

MANAGEMENT OF INTELLECTUAL PROPERTY RIGHTS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the benefits of IPR management ? (CO3, CO4)

   (b) Explain the situations where valuation of IP is important for an organization. (CO3, CO4)

   (c) Elaborate the general principles of granting patents as proceeded under the Patents Act, 1970. (CO3, CO4)

2. (a) Explain the key focus areas of Intellectual Property Audit. (CO2, CO5)

   (b) What are the benefits of a patent portfolio ? List down the contents of a patent portfolio. (CO2, CO5)

   (c) Describe in brief : exclusive, non-exclusive and sole license. (CO2, CO5)

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3. (a) What are the basic elements of patentability ? Discuss. (CO5, CO6)

   (b) Define 'Inventive step' under the Indian Patent Act. List down the various types of Patents. (CO5, CO6)

   (c) Explain the different factors affecting IP Management behaviour. Mention the attributes of a good IP Manager. (CO5, CO6)

4. (a) Who are the persons entitled to file a patent application ? Explain. (CO4, CO5)

   (b) Discuss the process of drafting a patent specification. (CO4, CO5)

   (c) Discuss in brief the historical evolution of IPR laws under International Regime. (CO4, CO5)

5. (a) Write a note on the salient features of Designs Act, 2000. Discuss the essential requirements for the registration of 'design' under the Designs Act, 2000. (CO1, CO2)

   (b) What is Utility Model ? Discuss the benefits of the rights acquired under the Utility Model Protection. (CO1, CO2)

   (c) Distinguish between Copyright and Industrial Design. (CO1, CO2)



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Roll No. ....ME 6th...........

# DEEM–05

## B. TECH. (SIXTH SEMESTER)

## END SEMESTER EXAMINATION, June, 2023

OPERATION RESEARCH

**Time : Three Hours**

**Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1\. (a) Use the graphical method top solve the following LP problem.

Maximize :

$$Z = 15x_1 + 10x_2$$

subject to the constraints :

(i) $4x_1 + 6x_2 \leq 360$

(ii) $3x_1 + 0x_2 \leq 180$

(iii) $0x_1 + 5x_2 \leq 200$

and $x_1, x_2 \geq 0$. (CO1)

(b) Solve the following LP problems using the simple method : (CO1)

Max. :

$$Z = 5x_1 + 3x_2$$

subject to :

$$x_1 + x_2 \leq 2$$

$$5x_1 + 2x_2 \leq 10$$

$$3x_1 + 8x_2 \leq 12$$

and $x_1, x_2 \geq 0$.

(c) A diet is to contain at least 20 ounces of protein and 15 ounces of carbohydrate. There are three foods A, B and C available in the market, costing ₹ 2, ₹ 1 and ₹ 3 per unit, respectively. Each unit of A contains 2 ounces of protein and 4 ounces of carbohydrate. Each unit of B contains 3 ounces of protein and 2 ounces of carbohydrate; and each unit of C contains 4 ounces of protein and 2 ounces of carbohydrate. How many units of each food should the diet contain so that the cost per unit diet is minimum ? (CO1)

2. (a) A company has four warehouses, a, b, c and d. It is required to deliver a product from these warehouses to three customers A, B and C. The warehouses have the following amounts in stock : (CO2)

| Warehouse | No. of units |
|---|---|
| a | 15 |
| b | 16 |
| c | 12 |
| d | 13 |

and the customer's requirements are

| Customer | No. of units |
|---|---|
| A | 18 |
| B | 20 |
| C | 18 |

The table below shows the costs of transporting one unit from warehouse to the customer :

Warehouse

| | | a | b | c | d |
|---|---|---|---|---|---|
| Customer | A | 8 | 9 | 6 | 3 |
| | B | 6 | 11 | 5 | 10 |
| | C | 3 | 8 | 7 | 9 |

Find the optimal transportation routes.

(b) A departmental head has four subordinates and four tasks to be performed. The subordinates differ in efficiency and the tasks differ in their intrinsic difficulty. His estimates of the times that each would take to perform each task is given in the matrix below : (CO2)

Tasks

| | | I | II | III | IV |
|---|---|---|---|---|---|
| Subordinates | A | 8 | 26 | 17 | 11 |
| | B | 13 | 28 | 4 | 26 |
| | C | 38 | 19 | 18 | 15 |
| | D | 19 | 26 | 24 | 10 |

How should the taske be allocated to subordinates so as to minimize the total man-hours ?

(c) ABC Tool Company has a sales force of 25 men, who operate from three regional offices. The company produces four basic product lines of hand tools. Mr. Jain, the sales manager, feels that 6 salesmen are needed to distribute product line I, 10 to distribute product line II, 4 for product line III and 5 salesmen for product line IV. The cost (in ₹) per day of assigning salesman from each of the offices for selling each of the product lines are as follows : (CO2)

| | | Product Lines | | | |
|---|---|---|---|---|---|
| | | I | II | III | IV |
| | A | 20 | 21 | 16 | 18 |
| Regional Office | B | 17 | 28 | 14 | 16 |
| | C | 29 | 23 | 19 | 20 |

At the present time, 10 salesman are allocated to office A, 9 to office B and 7 salesmen to office C. How many salesmen should be assigned from each office to sell each product line in order to minimize costs ?

3. (a) A research and development department is developing a new power supply for a console television set. It has broken the job down into the following : (CO3, CO4)

| Job | Description | Immediate Predecessors | Time (days) |
|---|---|---|---|
| A | Determine output voltage | – | 5 |
| B | Determine whether to use solid state rectifiers | A | 7 |

| | | | |
|---|---|---|---|
| C | Choose rectifier | B | 2 |
| D | Choose filters | B | 3 |
| E | Choose transformer | C | 1 |
| F | Choose chassis | D | 2 |
| G | Choose rectifier mounting | C | 1 |
| H | Layout chassis | E, F, | 3 |
| I | Build and test | G, H | 10 |

(i) Draw the network diagram of activities involved in the project and indicate the critical path.

(ii) What is the minimum completion time for the project ?

(b) A small project consists of seven activities, the details of which are given below : (CO3, CO4)

| Activity | Duration (days) | | | Immediate Predecessor |
|---|---|---|---|---|
| | Most Likely | Optimistic | Pessimistic | |
| A | 3 | 1 | 7 | – |
| B | 6 | 2 | 14 | A |
| C | 3 | 3 | 3 | A |
| D | 10 | 4 | 22 | B, C |
| E | 7 | 3 | 15 | B |
| F | 5 | 2 | 14 | D, E |
| G | 4 | 4 | 4 | D |

(i) Draw the network, number the nodes, find the critical and path, the expected project completion time and the next most critical path.

(ii) What project duration will have 95 percent confidence of completion ?

(c) The required data for a small project consisting of different activities are given below : (CO3, CO4)

| Activity | Predecessor Activities | Normal | | Crash | |
|---|---|---|---|---|---|
| | | Duration (days) | Cost (₹) | Duration (days) | Cost (₹) |
| A | – | 6 | 300 | 5 | 400 |
| B | – | 8 | 400 | 6 | 600 |
| C | A | 7 | 4,00 | 5 | 600 |
| D | B | 12 | 1,000 | 4 | 1,400 |
| E | C | 8 | 800 | 8 | 800 |
| F | B | 7 | 400 | 6 | 500 |
| G | D, F | 5 | 1,000 | 3 | 1,400 |
| H | F | 8 | 500 | 5 | 700 |

(i) Draw the network diagram for the project and find the normal and minimum project length.

(ii) If the project is to be completed in 21 days with minimum crash cost which activities should be crashed to how many days ?

4. (a) The following matrix gives the payoff (in ₹) of different strategies (alternatives) $S_1$, $S_2$ and $S_3$ against condition (events) $N_1$, $N_2$, $N_3$ and $N_4$. (CO5)

| Strategy | State of Nature | | | |
|---|---|---|---|---|
| | $N_1$ | $N_2$ | $N_3$ | $N_4$ |
| $S_1$ | 4,000 | –100 | 6,000 | 18,000 |
| $S_2$ | 20,000 | 5,000 | 400 | 0 |
| $S_3$ | 20,000 | 15,000 | –2,000 | 1,000 |

Indicate the decision-taken under the following approaches :

(i) Pessimistic

(ii) Optimistic

(iii) Equal probability

(iv) Regret

(v) Hurwicz criterion, the degree of optimism being 0.7.

(b) Explain the principle of dominance in game theory and solve the following game : (CO5)

**Player B**

| Player A | $B_1$ | $B_2$ | $B_3$ | $B_4$ | $B_5$ |
|---|---|---|---|---|---|
| $A_1$ | 2 | 4 | 3 | 8 | 4 |
| $A_2$ | 5 | 6 | 3 | 7 | 8 |
| $A_3$ | 6 | 7 | 9 | 8 | 7 |
| $A_4$ | 4 | 2 | 8 | 4 | 3 |

(c) You are given the following payoffs of three acts $A_1$, $A_2$ and $A_3$ and the events $E_1$, $E_2$, $E_3$ : (CO5)

| States of Nature | Three Acts | | |
|---|---|---|---|
| | $A_1$ | $A_3$ | $A_3$ |
| $E_1$ | 25 | – 10 | – 125 |
| $E_2$ | 400 | 440 | 400 |
| $E_3$ | 650 | 740 | 750 |

The probabilities of the states of nature are 0.1, 0.7 and 0.2, respectively. Calculate the tabulate the EMV and conclude which would prove to be the best course of action.

5. (a) We have six jobs, each of which must go through machines A, B and C in the order ABC. Processing time (in hours) are given in the following table : (CO6)

| Job | Machine A | Machine B | Machine C |
|---|---|---|---|
| 1 | 8 | 3 | 8 |
| 2 | 3 | 4 | 7 |
| 3 | 7 | 5 | 6 |
| 4 | 2 | 2 | 9 |
| 5 | 5 | 1 | 10 |
| 6 | 1 | 6 | 9 |

Determine a sequence for the five jobs that will minimize the elapsed time $t$.

(b) Find an optimal sequence for the following sequencing problem of four jobs and five machines (when passing is not allowed) of which processing time (in hrs) is as follows : (CO6)

| **Job** | **Machine $M_1$** | **Machine $M_2$** | **Machine $M_3$** | **Machine $M_4$** | **Machine $M_5$** |
|---|---|---|---|---|---|
| 1 | 6 | 4 | 1 | 2 | 8 |
| 2 | 5 | 5 | 3 | 4 | 9 |
| 3 | 4 | 3 | 4 | 5 | 7 |
| 4 | 7 | 2 | 2 | 1 | 5 |

Also find the total elapsed time.

(c) Use the graphical method to minimze the time needed to process the following jobs on the machines shown, i. e. for each machine find the job that should be done first. Also, calculate the total elapsed time to complete both jobs : (CO6)

| | **Machine** | | | | |
|---|---|---|---|---|---|
| | A | B | C | D | E |
| Job 1 $\begin{cases} Sequence \\ Time\,(hrs) \end{cases}$ | 3 | 4 | 2 | 6 | 2 |

| | Machine | | | | |
|---|---|---|---|---|---|
| Job 2 $\begin{cases} Sequence \\ Time\,(hrs) \end{cases}$ | B | C | A | D | E |
| | 5 | 4 | 3 | 2 | 6 |